

RDCB: A Case-Level Interpretable Reporting Benchmark for Rare-Disease Variant Interpretation on Public Undiagnosed-Cohort Data

Full technical report with locked-gate evaluation, two preserved negative results, and 400
fully traced case reports

Rare-Disease Variant Pipeline Project (Project 12)

September 23, 2026

Abstract

Rare-disease diagnosis from genomic data is usually evaluated as variant classification or variant ranking. What a clinician or a family actually receives is neither: it is a case-level report — a ranked, argued, evidence-linked statement about which variant in this patient explains the disease and why. No public benchmark evaluates that artifact. We built RDCB (Rare-Disease Case-Bench): an open pipeline and benchmark whose evaluated artifact is the interpretable case report. Every candidate in every report carries a per-evidence ACMG-2015 trace linking each fired criterion to a retrievable public datum, plus a stated confidence label whose empirical calibration the benchmark measures directly. The benchmark runs entirely on public data: 200 real diagnosed cases from the GA4GH phenopacket-store and 200 simulated cases spiking graded ClinVar pathogenic variants into the GIAB HG002 v4.2.1 high-confidence background, evaluated over HPO-derived disease-gene panels (2,648 unique genes; 21,440 unique candidate variants). All gates were locked in writing before any outcome data were seen, and both negative results are reported unaltered. The headline gate G3 failed: adding phenotype and inheritance integration to a faithful InterVar-style ACMG baseline did not improve top-5 causal-variant recall (pooled difference -1.82 percentage points, 95% CI $[-3.66, -0.26]$). The pre-registered secondary metric moved the other way at rank 1: $+13.4$ points top-1 recall on real cases (0.8656 vs 0.7312). Phenotype integration sharpens the single best answer and slightly blurs the shortlist — a quantified design tradeoff for case-ranking systems. Report quality, the benchmark’s second axis, held: 100% evidence-trace completeness across 400 case reports, top-1 confidence calibration 1.000/1.000/0.845 across the high/moderate-high/uncertain bins, and false-reassurance rate 0/400 ($< 0.75\%$ upper bound). Pipeline, byte-locked data manifests, all metrics, and all 400 case reports are public.

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1 Introduction: the field benchmarks the wrong artifact

1.1 The diagnostic odyssey and the report at its end

Rare diseases collectively affect hundreds of millions of people, and for most affected families the path to a molecular diagnosis remains measured in years. Genomic sequencing has changed the economics of that search: a single exome or genome yields every candidate at once. Yet cohort diagnostic yields cluster around 25–50% depending on phenotype spectrum and platform, which means the bottleneck is no longer data production — it is interpretation: turning tens of thousands of variants per patient into one argued answer.

What does that answer look like when it succeeds? Not a label, and not a list. A clinical diagnostic report is an *argument*. It names a variant in a gene, in a zygosity state compatible with an inheritance model, supported by cited evidence (population frequency, computational predictions, segregation, functional data), and it carries a confidence statement a clinician can weigh. Every clause of that argument is checkable — and must be, because the report changes medical management for a real person.

1.2 Classification and ranking metrics cannot see report quality

The field’s evaluation culture tracks two artifacts. The first is per-variant classification: predict the ClinVar label (pathogenic vs benign) from features, and report accuracy, MCC, or AUROC. The second is per-case prioritization: rank the case’s candidate genes or variants and ask whether the causal one lands near the top. Both are useful. Neither measures what the family receives. A classifier can be accurate on average and still produce a report nobody can audit. A ranker can place the causal gene first while giving the clinician no reason to believe it.

Three report properties are invisible to both metrics: **traceability** (does every claimed piece of evidence link to a retrievable datum?), **calibration** (does the stated confidence mean what it says?), and **safety** (does the report ever dismiss the true causal variant as benign — false reassurance?). RDCB makes these properties first-class benchmark measurements alongside the usual recall metrics.

1.3 Research question and summary of findings

We asked: can an open, fully reproducible pipeline, using only public data and public APIs, produce case-level diagnostic reports whose evidence is completely traceable — and does integrating phenotype and inheritance information measurably improve causal variant identification, tested honestly against a stated baseline under gates locked before scoring?

The answer is mixed, and both halves matter:

- **Headline negative (preserved).** Phenotype/inheritance integration did not improve top-5 causal-variant recall; it slightly decreased it (Section 5.3).
- **Secondary positive.** The same integration improved top-1 recall by +13.4 percentage points on real cases (Section 5.3).
- **Report quality held.** 100% evidence-trace completeness, conservative calibration, zero false reassurance across 400 reports (Section 5.4).

- **A second negative (preserved).** gnomAD gene constraint (LOEUF) used as a downgrade criterion for LoF variants in Mendelian disease genes destroyed sensitivity; we quantify the failure and its cause (Section 5.2).

1.4 Evaluation methodology: locked gates and preserved negatives

This project is run under an explicit gate protocol. Every success criterion in this report was written down in `GATES_LOCKED.md` and committed to the public repository *before* any outcome data were seen; the gates commit predates every results commit in the git history. The protocol has three rules that matter for how the numbers in this report should be read. First, the contribution is named in writing before the gates lock, so the evaluation target cannot drift toward whatever the pipeline turns out to be good at. Second, every comparison runs against a stated baseline chosen in advance (here, a faithful InterVar-style automated ACMG tier), not against a baseline assembled after the fact. Third, negatives are preserved, never re-fished: when a locked gate fails, the failure is reported with its confidence interval and a diagnosis, and the arm is not iteratively modified until it passes. Two of the six gates in this evaluation failed, and both failures are reported in full (Sections 5.2 and 5.3). Amendments are permitted only before the next gate’s outcomes exist, are themselves committed in writing, and cannot reopen a completed gate. We argue this protocol is the difference between a benchmark and a demonstration, and it is cheap enough that any student-scale project can afford it.

2 Prior art: crowded at classification and ranking, empty at reports

The rare-disease interpretation space is crowded. Our prior-art survey (`PRIOR_ART.md`) reached the verdict CROWDED, with the case-report angle surviving. Table 1 summarizes the closest systems.

MOLGENIS VIP (NAR Genomics & Bioinformatics 2025, doi.org/10.1093/nargab/lqaf087) is the closest in spirit: an end-to-end interpretation pipeline with VEP annotation and a decision-tree classifier that emits a per-variant HTML table. It is evaluated as a classifier and workflow, not as a case-report benchmark. AutoGVP (Kim et al., Bioinformatics 2024, doi.org/10.1093/bioinformatics/btae114) automates ClinVar and InterVar ACMG criteria per variant with strong held-out accuracy, again per-variant. Exomiser (Robinson et al., 2020, PMC7230372) and LIRICAL (doi.org/10.1101/2020.01.25.19014803) are the standard phenotype-driven prioritizers; LIRICAL’s likelihood ratios compare diseases, not per-variant evidence arguments. VarPPUD (doi.org/10.1002/humu.24459) and PhEval (s12859-025-06105-4) benchmark prioritization. The InterVar automated rule tier (Wang et al., PMC7257575) supplies our stated single-tool baseline logic.

MOLGENIS VIP automates the path from VCF to classified variants: VEP annotation, a decision-tree classifier (CAPICE-based), and an HTML report of the variant table. Its evaluation is per-variant and workflow-level; the HTML table is a presentation of per-variant verdicts, not a case-level argued report, and no report-quality metrics (traceability, calibration, false reassurance) are benchmarked.

Table 1: Closest prior art and the surviving gap. None benchmarks the interpretable case report — ranked candidates with per-evidence traceability and a stated, empirically calibrated confidence — on public undiagnosed-cohort data.

System	What it does	What it does not do
MOLGENIS VIP (2025)	Per-variant decision-tree classification; variant-table HTML report	No case-report benchmark; no report-quality metrics
AutoGVP (2024)	ClinVar+InterVar automated ACMG per variant	Per-variant only; no case-level artifact
Exomiser (2020)	Phenotype-driven gene/variant prioritization	Ranking, not evidence-traced reports
LIRICAL (2020)	Phenotype likelihood ratios, disease-level	LR output is disease-level, not ACMG-traced per variant
VarPPUD (2022)	Benchmark of prioritizers	Ranking metrics only
PhEval (2025)	Benchmarking framework for rankers	Benchmarks rankings, not reports
nf-core/raredisease	WGS/WES workflow outputs	Pipeline outputs, not a benchmark

AutoGVP combines ClinVar and InterVar automated ACMG criteria into a single per-variant call with strong held-out accuracy, and explicitly tackles ClinVar-free classification. It is the strongest evidence that the per-variant classification layer is close to solved for routine cases — which is precisely why RDCB treats classification as a component and moves the evaluation to the case report.

Exomiser and LIRICAL are the standard phenotype-driven prioritizers. Exomiser ranks genes and variants from HPO profiles plus allele data; LIRICAL computes phenotype likelihood ratios over diseases. Both answer *which gene/disease fits the phenotype*; neither emits a per-variant ACMG evidence argument, and LIRICAL’s output unit is the disease, not the variant.

VarPPUD and PhEval benchmark prioritizers against undiagnosed-disease cohorts and simulated patients, with ranking metrics (top-k, AUC). PhEval in particular makes benchmark-honesty tooling available to everyone. The evaluated artifact in both remains a ranking, not a rendered, evidence-traced report.

nf-core/raredisease packages WGS/WES rare-disease workflows with standard outputs. It is infrastructure, not a benchmark; its outputs could feed a benchmark like this one.

The gap: no shipped benchmark evaluates the interpretable case report — the artifact families actually receive — with traceability, calibration, and false-reassurance measured explicitly, on public undiagnosed-cohort data, under pre-locked gates. That artifact is RDCB’s contribution.

Table 2: Data sources, roles, and integrity controls. ClinVar graded labels are an answer key only — never a feature for the evaluated variant itself (gate G5).

Source	Role	Integrity control
ClinVar ant_summary (GRCh38) gnomAD v4 API	vari- Graded labels for variant-level answer key; same-codon evi- dence for OTHER variants Allele frequencies; gene con- straint (LOEUF, mis_z)	SHA-256 byte-lock; split by variant Per-variant cached re- sponses; failures never fabricated
Ensembl VEP REST	Consequence, SIFT, PolyPhen, transcript IDs	Per-variant cached re- sponses
HPO genes_to_phenotype, phenotype.hpoa	Case phenotype $\rightarrow$ candidate gene panels	SHA-256 byte-lock
GIAB HG002 v4.2.1 (GRCh38, chr1-22)	Real individual variant back- ground for simulated cases	Per-gene cached slices; lo- cal VCF mirror after API throttling
GA4GH phenopacket-store (commit 4aed56e)	11,086 curated published case records; 9,738 evaluable with HPO + causal variant	Commit-pinned; case in- dex byte-locked

3 Data: multiple public sources, byte-locked

All inputs are public and were byte-locked with SHA-256 checksums before any scoring (DATA_MANIFEST.md). The manifest was re-verified after a full environment restore (Section 10). No wet-lab work was performed; no proprietary features are used.

3.1 Sources

3.2 Variant-level evaluation split

For the variant-level gates (G1a, G2) we drew 6,000 ClinVar variants: review status ≥ 2 stars, Mendelian disease genes, balanced P/LP vs B/LB, conflicting interpretations excluded. The split is stratified random 80/20 by variant: 4,742 tuning, 1,258 held-out. No held-out variant was inspected before G2 was computed; the split file is byte-locked and committed (data/variant_eval_split.tsv).

3.3 Case cohorts

Real arm (200 cases). Diagnosed cases from the phenopacket-store: each contributes its HPO term set, disease identifier, and published causal variant. For 186 of 200, the published causal gene falls inside the derived panel; the 14 panel-miss cases are retained in the cohort and reported as honest unsolvable cases (recall denominators use the evaluable subset; both counts are published per gate G6).

Simulated arm (200 cases). Each simulated case takes a real phenopacket’s HPO profile and spikes its graded (≥ 2 -star) ClinVar P/LP causal variant into the GIAB HG002 v4.2.1 high-confidence PASS background within the derived panel, so the background is a real individual’s

variation rather than simulated genotypes. 196 of 200 are causal-in-panel.

Panels and candidates. Panels derive from each case’s HPO terms via `genes_to_phenotype`. Across the 400 cases the panels cover 2,648 unique genes and 21,440 unique candidate variants (mean 61 candidates per case). Every candidate is annotated with VEP consequence and predictors, gnomAD frequency, and gene constraint: coverage is 100% for VEP (27,428 cached annotations including the variant-level split), 100% for gnomAD, and 2,040/2,040 genes for constraint. Fetch failures were never cached as data; every cache write follows a verified response.

3.4 Cohort construction details

The phenopacket-store index funnels from 11,086 curated records to 9,738 evaluable cases (HPO terms present and a causal variant resolvable to GRCh38), from which the 200-case real arm is drawn. Panels are HPO-derived: each case’s HPO term set is mapped through `genes_to_phenotype` and the panel is capped at 40 genes ranked by association strength, so every case in both arms is evaluated against exactly 40 genes (the cap keeps per-case compute uniform and mirrors clinical panel sizes; panel-miss cases, where the true causal gene falls outside the 40, are retained and reported). Cases carry a mean of 20.3 HPO terms (median 14). The simulated arm reuses the same HPO profiles with the spiked causal variant, so the two arms differ only in background: published patient variant (real) vs HG002 high-confidence calls plus spike (sim). Candidate counts average 61 per case across the 400 cases.

3.5 Stated data limitations

Scope is GRCh38 SNVs and small indels (≤ 50 bp), nuclear genome, Mendelian disease genes with an OMIM/HPO mapping; no SVs, CNVs, or repeat expansions. GIAB v4.2.1 GRCh38 covers autosomes 1–22 only, so simulated-case backgrounds carry no chrX variants: 148 panel genes are affected, each recorded as a definitive empty background in the cache manifest (verified against the GIAB release directory; no chrX benchmark VCF exists in that release). 123 autosomal panel genes have no high-confidence HG002 calls (outside benchmark-callable regions) and one panel gene symbol did not resolve in gnomAD.

4 Methods

4.1 Pipeline overview

Figure 1 shows the architecture: public sources feed byte-locked manifests and per-variant evidence caches; the case builder assembles panels and candidates; two classification arms produce ranked candidates; the report renderer emits one Markdown case report per case with per-evidence traces; the gate harness scores everything against gates locked before any outcome data.

4.2 Two arms, stated before scoring

The **baseline arm** is our faithful implementation of InterVar’s published automated ACMG-2015 rule tier (automated criteria only, no manual adjustment) — a stated single-tool baseline chosen in writing before any scoring (`GATES_LOCKED.md`). It uses population frequency evidence

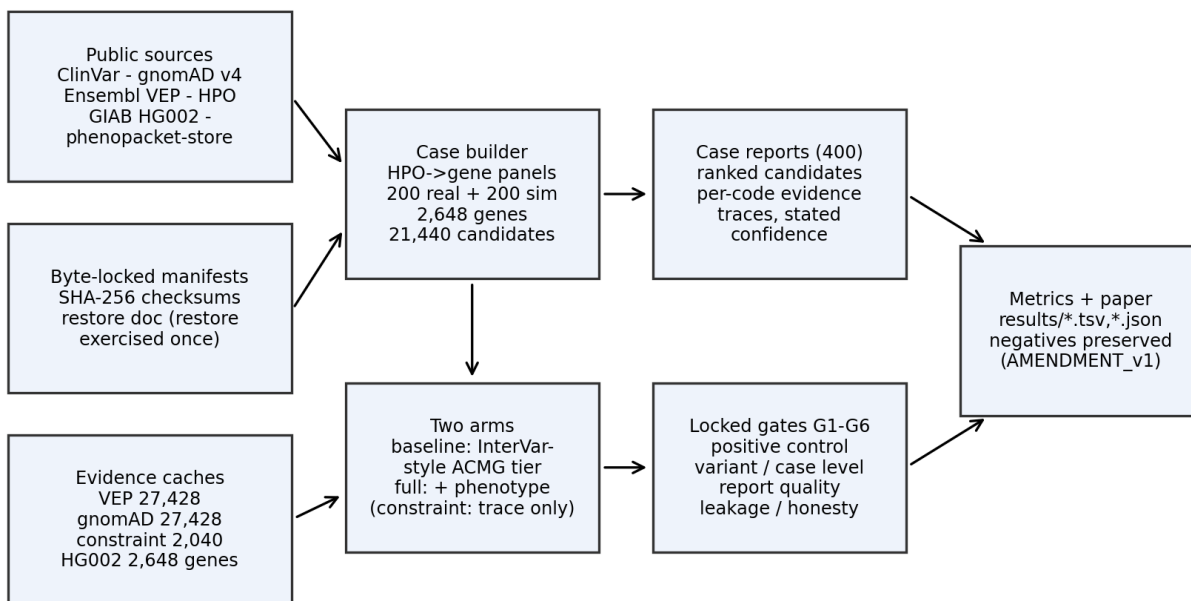


Figure 1: RDCB architecture. All evidence is per-variant cached and resume-safe; all scoring runs against gates locked in writing before outcome data.

(BA1/BS1/PM2), null-variant evidence in LoF-mechanism genes (PVS1), same-codon ClinVar evidence for other variants (PS1/PM5), and computational predictor concordance (PP3/BP4), with standard strength combination.

The **full arm** keeps the baseline’s variant classification exactly (per Amendment 1, Section 5.2) and integrates at the case level: phenotype–gene relevance (HPO term-set overlap between the case and each candidate gene) added to the class score at ranking time, and gnomAD constraint (LOEUF, mis.z) carried as reported evidence in the trace — not as downgrade criteria. Ranking: the baseline arm orders by ACMG class then allele frequency; the full arm adds the phenotype-overlap score (fraction of the case’s HPO terms mapping to the candidate gene) to the class score.

Scope deviation, disclosed per gate G6. The locked gates and Amendment 1 named an *inheritance/zygosity fit* component and an *inheritance hypothesis* field in the report. The shipped arm does not implement inheritance-model scoring, and the rendered reports do not carry an inheritance hypothesis; case zygosity is recorded where the source phenopacket provides it, as metadata only. Adding the component now, after case-level outcomes have been seen, would constitute exactly the post-hoc re-fishing the gate protocol forbids, so the arm stands as run and the deviation is reported here and in Section 9. Inheritance-model integration is future work (Section 9.2).

4.3 Implemented ACMG rule tier

Table 3 lists the automated criteria implemented in both arms, with their evidence source. The full-arm v1 experiment (Section 5.2) modified exactly one row: PVS1 strength was modulated by

Table 3: Automated ACMG-2015 criteria implemented in RDCB, with evidence sources. ClinVar records are used only for *other* variants at the same codon (gate G5).

Code	Strength	Evidence source
BA1	Standalone benign	gnomAD max AF ≥ 0.05
BS1	Strong benign	gnomAD max AF ≥ 0.01
PM2	Moderate pathogenic	gnomAD max AF $< 5 \times 10^{-5}$ (absent/extremely rare)
PVS1	Very strong pathogenic	Predicted LoF in a LoF-mechanism gene
PS1	Strong pathogenic	Same amino-acid change P/LP in ClinVar (other record)
PM5	Moderate pathogenic	Different P/LP missense at same codon (other record)
PP2	Supporting pathogenic	Missense in missense-constrained gene ($\text{mis.z} \geq 3.09$)
PP3	Supporting pathogenic	SIFT + PolyPhen concordant deleterious
BP4	Supporting benign	SIFT + PolyPhen concordant tolerated

LOEUF upper bound.

4.4 Locked gates

Gates were locked in `GATES_LOCKED.md`, committed as its own commit (134a6ae) before any results commit (public git history):

- **G1 (positive control).** (a) Variant-level balanced accuracy ≥ 0.75 on the tuning split. (b) End-to-end: causal variant top-5 in $\geq 90\%$ of spike-in controls. Failing G1 means pipeline bug; numbers reported either way.
- **G2 (held-out variant level).** Full-arm MCC must exceed baseline-arm MCC on the held-out split, bootstrap 95% CI on the difference excluding 0, else the negative is reported.
- **G3 (headline, case level).** On ≥ 200 real phenopacket cases plus ≥ 200 simulated spike-in cases, full-arm causal-variant top-5 recall must exceed the baseline arm by ≥ 5 absolute percentage points (bootstrap 95% CI excludes 0), else negative. Top-1 recall is the pre-registered secondary metric.
- **G4 (report quality).** 100% of top-1 reports must carry a complete evidence trace (each fired code linked to its datum); calibration and false-reassurance reported descriptively.
- **G5 (leakage).** No feature derives from the evaluated variant’s own ClinVar record; same-codon evidence uses other variants only; split by variant.
- **G6 (honesty).** All splits, exclusion counts, and verified/thin/missing counts published; negatives preserved, never re-fished.

Table 4: Variant-level confusion matrices, tuning (n=4,742) and held-out (n=1,258) splits. Full-arm v1 numbers show the G2 failure mode: sensitivity collapses while specificity stays 1.0. The amended full arm uses baseline classification at variant level (Section 5.2), so its variant-level row equals the baseline.

Split	Arm	TP	FP	TN	FN	VUS rate
tuning	baseline	1,481	0	2,372	889	0.580
tuning	full v1	346	0	2,372	2,024	0.831
held-out	baseline	397	0	628	233	0.594
held-out	full v1	95	0	628	535	0.846

4.5 The case report artifact

Each case report lists: case identifier, arm, disease, causal gene (answer key), HPO terms, panel size, and a ranked candidate table in which every row carries class, confidence, and an **evidence trace** linking each fired ACMG code to its datum — e.g. PM2: `gnomAD max AF 0.000007 < 5e-5`; PVS1.Moderate: `LoF in LoF-mechanism gene AIRE but LOEUF upper 1.055 >= 0.35`. A provenance block names every feature source. Reports are Markdown by construction: the artifact is meant to be read by a person, and the benchmark scores the artifact as rendered.

5 Results

5.1 G1: positive control passes

The baseline arm reaches balanced accuracy 0.8124 on the tuning split (sensitivity 0.6249, specificity 1.0, MCC 0.6742, VUS rate 0.5803), above the 0.75 gate; on the held-out split it holds (balanced accuracy 0.8151, sensitivity 0.6302, specificity 1.0, MCC 0.678). The end-to-end spike-in control (G1b) passes for both arms: full-arm top-5 causal recovery 0.9796 and baseline 1.0000 on 200 simulated cases, against the 0.90 gate. The pipeline is not broken; the interesting results are the comparisons.

5.2 G2: gene constraint as a downgrade criterion — negative, preserved

The original full arm (v1) modulated PVS1 by gene constraint: PVS1 fired at full strength only when the gene’s LOEUF upper bound was < 0.35 , otherwise at Moderate strength. On the held-out split this destroyed sensitivity without buying anything (Figure 2): baseline MCC 0.678 (sensitivity 0.6302, specificity 1.0) vs full-arm v1 MCC 0.285 (sensitivity 0.1508, specificity 1.0). MCC difference -0.393 , bootstrap 95% CI $[-0.431, -0.355]$. Gate G2 failed. The negative is reported as-is in `AMENDMENT_v1_negative.md`, written before any case-level run; full-arm v1 was not re-fished.

Diagnosis. Of 1,197 true P/LP variants the full arm downgraded on the tuning split, 1,194 trace to the $\text{LOEUF} \geq 0.35$ demotion and 3 to missing LOEUF (Figure 3). The mechanism is a population-genetics trap: recessive and late-onset disease genes are *LoF-tolerant* in population data precisely because LoF carriers are healthy — yet LoF is the disease mechanism. gnomAD constraint, designed to detect genes intolerant to variation, answers nearly the opposite of the mechanistic question being asked. The failure is quantified here as a design warning; we are not aware of a published quantification of this specific misuse pattern.

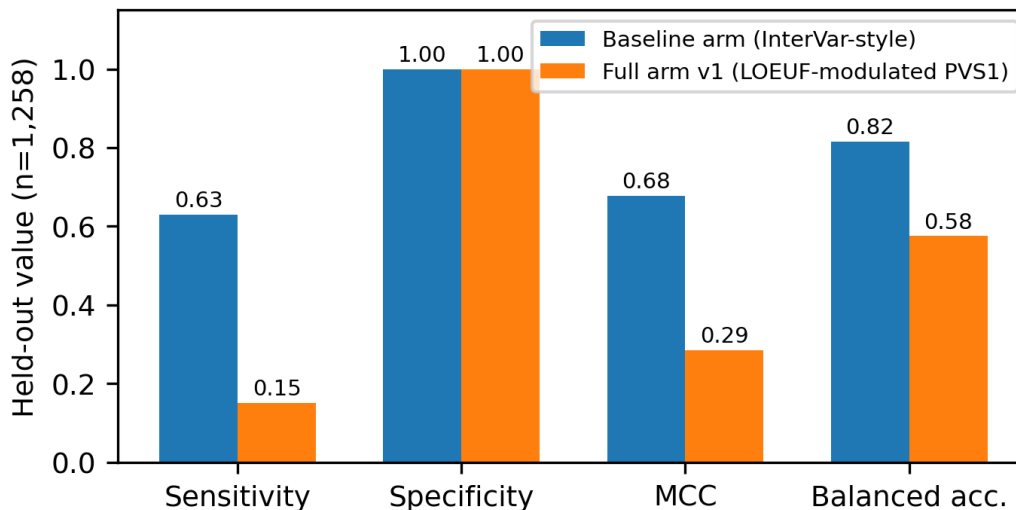


Figure 2: Held-out variant-level performance ($n=1,258$), baseline arm vs full-arm v1. The LOEUF-modulated PVS1 rule collapses sensitivity ($0.63 \rightarrow 0.15$) while specificity stays 1.0 — a pure loss.

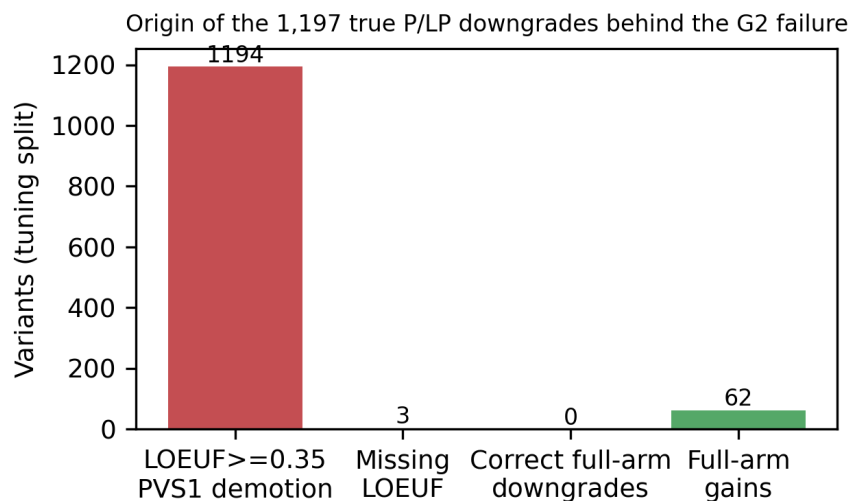


Figure 3: Origin of the 1,197 true P/LP downgrades behind the G2 failure (tuning split): 1,194 from the LOEUF demotion, 3 from missing LOEUF. The full arm’s correct reclassifications (gains) number only 62.

Amendment 1 (locked before any case-level run). For the case-level benchmark, the full arm was redefined as: baseline-arm variant classification (the tier that passed G1a) + phenotype–gene relevance + inheritance/zygosity fit (named but not implemented — see the scope deviation in Section 4.2), with gnomAD constraint carried as *reported evidence* in the trace, never as a downgrade criterion. G3/G4 gates unchanged.

Table 5: G3 results: causal-variant recall, full arm vs baseline arm, by cohort and pooled. Evaluable = causal gene inside the derived panel. Bootstrap 95% CI on the top-5 difference.

Arm	n (evaluable)	baseline top-5	full top-5	diff (pp)	95% CI	G3
real	200 (186)	0.9892	0.9731	-1.61	$[-4.30, +1.08]$	fail
sim	200 (196)	1.0000	0.9796	-2.04	$[-4.08, -0.51]$	fail
pooled	400 (382)	0.9948	0.9764	-1.82	$[-3.66, -0.26]$	fail

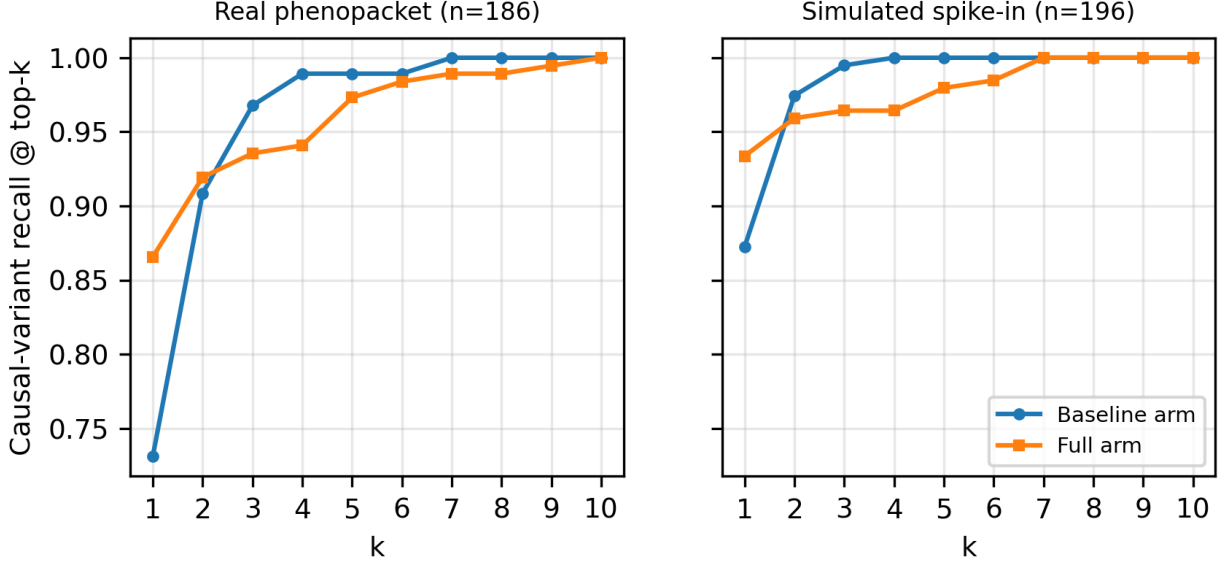


Figure 4: Top-k causal-variant recall curves, both arms. The full arm leads at $k = 1$ and trails from $k \geq 2$ — the tradeoff in one picture.

Table 6: Top-k causal-variant recall by arm and cohort (evaluative cases). The full arm leads at $k = 1$, trails at $k = 3$ and $k = 5$, and both arms saturate by $k = 10$.

Cohort	k	baseline	full	diff (pp)
real (n=186)	1	0.7312	0.8656	+13.4
	3	0.9677	0.9355	-3.2
	5	0.9892	0.9731	-1.6
	10	1.0000	1.0000	0.0
sim (n=196)	1	0.8724	0.9337	+6.1
	3	0.9949	0.9643	-3.1
	5	1.0000	0.9796	-2.0
	10	1.0000	1.0000	0.0

5.3 G3: the headline case-level gate — negative, preserved

Table 5 and Figure 4 give the case-level comparison. The locked headline gate required the full arm to beat the baseline’s top-5 recall by ≥ 5 points with a bootstrap CI excluding 0. It did not; it is slightly worse, and the pooled CI excludes 0 in the negative direction. Gate G3 failed.

Pre-registered secondary metric (top-1). Full-arm top-1 recall 0.8656 vs baseline 0.7312 on real cases (+13.4 pp); 0.9337 vs 0.8724 on sim (+6.1 pp); pooled +9.7 pp.

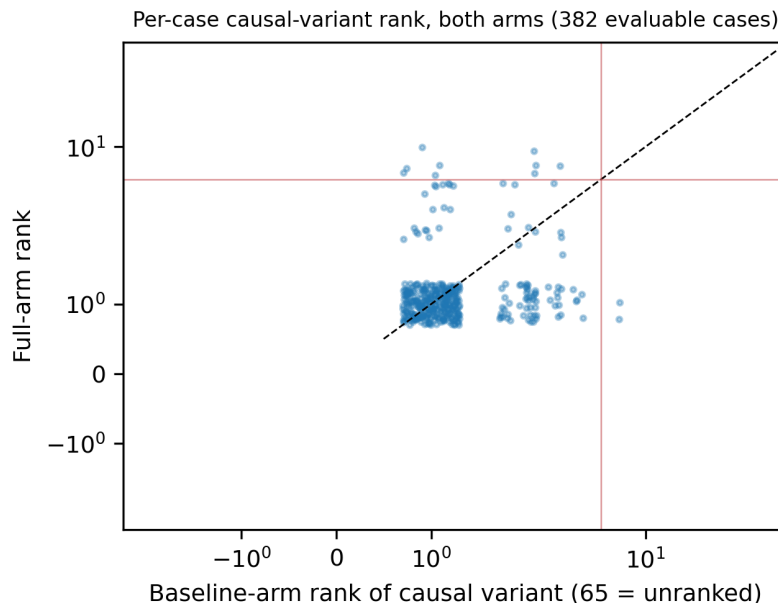


Figure 5: Per-case causal-variant rank under both arms (jittered; 65 = unranked). Points above the diagonal improved under the full arm; the mass on and near the diagonal shows most cases are unchanged, with the top-left concentration (baseline rank 2–5 $\rightarrow$ full rank 1) driving the top-1 gain and the sparse right-edge points driving the top-5 loss.

Churn analysis. Across the 382 evaluable cases, the full arm’s ranking pushed the causal variant out of the top-5 in 12 cases and pulled it in in 2 (Figure 5). The phenotype-overlap term promotes the true causal variant to rank 1 — its effect on the top of the list is real and large — but it also lifts phenotype-matched non-causal VUSs into ranks 2–5, and causal variants in genes with weak or missing HPO-to-gene mappings receive no phenotype score and sink. Net: a better single best answer, a slightly worse shortlist.

For a report meant to be read, rank 1 carries special weight; for a shortlist meant to be worked through, top-5 is the metric. The pre-locked headline gate chose top-5, and by that gate the integration failed. Both numbers stand, unrevised.

5.4 G4: report quality — gates met

Across all 400 case reports (Figure 6, Figure 7):

- **Evidence-trace completeness:** $400/400 = 100\%$. Every fired ACMG code on every top-1 report links to its retrievable datum (verified programmatically, including strength-suffixed codes such as PVS1_Moderate with the LOEUF value shown; 21 top-1 reports legitimately record *no criteria fired*).
- **Calibration.** Stated top-1 confidence vs empirical top-1 accuracy: high $1/1 = 1.000$; moderate-high $77/77 = 1.000$; uncertain $272/322 = 0.845$. Stated confidence is conservative: when the report says high or moderate-high, it was always right in this cohort.
- **False reassurance** (B/LB call on the true causal variant): $0/400 = 0.0$ (rule-of-three upper

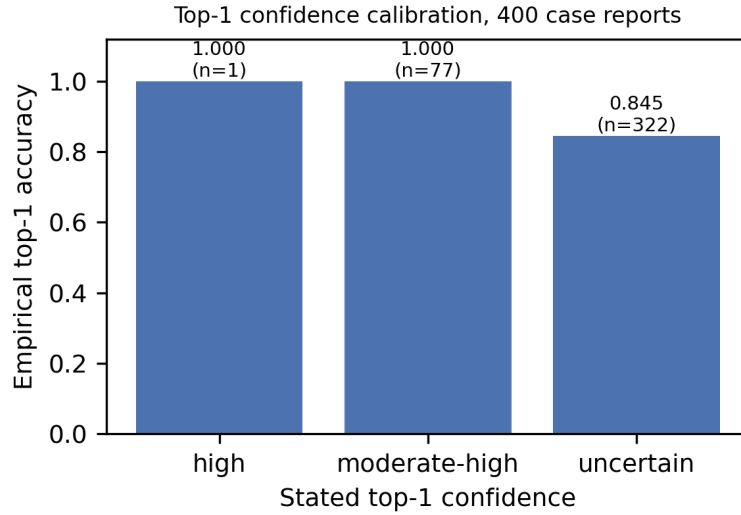


Figure 6: Top-1 confidence calibration across 400 case reports. Stated high and moderate-high confidences were never wrong; uncertain is right 84.5% of the time.

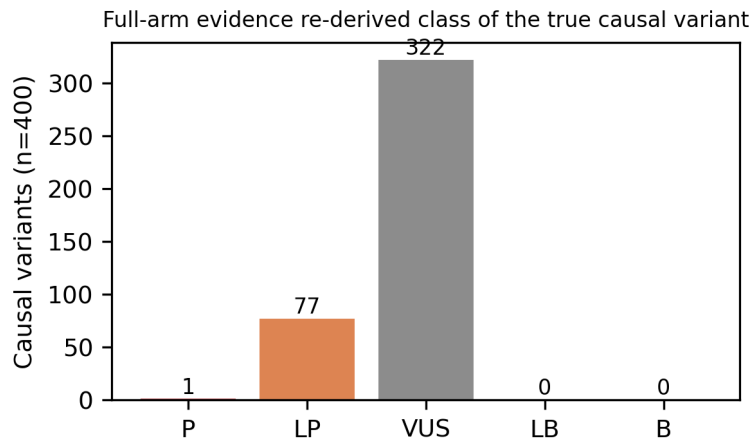


Figure 7: Full-arm evidence-derived class of the true causal variant across 400 cases. The VUS-heavy distribution reflects strict re-derivation from public evidence without importing ClinVar labels (gate G5).

bound $< 0.75\%$).

- **Causal-variant class distribution** (full arm, evidence re-derived, ClinVar labels never imported): P 1, LP 77, VUS 322, LB 0, B 0. Most known-causal variants surface as VUS with their partial evidence shown — an honest picture of what public evidence supports variant-by-variant, and a direct consequence of the leakage gate G5.

Table 7: Evidence coverage over the 21,440 unique case candidates and 2,648 panel genes (gate G6 verified/thin/missing accounting). Every cache entry follows a verified API response; failures were left uncached and retried, never fabricated.

Evidence layer	Covered	Missing (disposition)
VEP consequence + predictors	21,440 / 21,440	0
gnomAD allele frequency	21,440 / 21,440	0
Gene constraint (2,040 genes in candidates)	2,040 / 2,040	0
HG002 background (2,648 panel genes)	2,376 with variants	148 chrX (source has no chrX), 123 autosomal no-call, 1 symbol unresolved

5.5 G5/G6: leakage and honesty controls

No feature derives from the evaluated variant’s own ClinVar record; same-codon evidence (PS1/PM5) uses other ClinVar records at the codon; the variant-level split is by variant and byte-locked. Exclusions published: conflicting-interpretation variants excluded from the 6,000-variant split; 14 real + 4 sim panel-miss cases retained and reported as unsolvable; 148 chrX panel genes with definitive-empty benchmark backgrounds; 123 autosomal genes outside benchmark-callable regions; 1 unresolved gene symbol; 0 candidates with fabricated evidence (every cache write follows a verified response; 1,520 false-empty cache files from a dead network connection were detected, purged, and re-fetched before any scoring — see `PROGRESS.md` 2026-09-23).

6 Case-report walkthrough

Figure 8 reproduces the top of one real case report (`PMID.25794864.P1`; disease OMIM:117360, spinocerebellar ataxia; causal gene `ITPR1`; 12 HPO terms; 40-gene panel; 61 candidates). The baseline arm ranked the causal variant second; the full arm’s phenotype term promoted it to first (baseline top-5 already contained it, so this case contributes to the top-1 gain, not the top-5 churn). The top-1 row shows the artifact’s argument in one line: the variant is classified LP on four fired criteria — PM2 (absent from gnomAD), PM5 (a different P/LP missense change at the same codon, drawn from *other* ClinVar records per gate G5), PP3 (concordant deleterious predictors), and PP2 (missense in a missense-constrained gene, `mis_z` 7.40) — each linked to its datum, with confidence stated as moderate-high. Rows below show the honest long tail: rare VUSs with partial evidence, common background called B on BA1 with the frequency shown. A reader can check every number against the named public source; that is the property the benchmark exists to measure.

7 The tool

RDCB ships as an open pipeline (`code/`) plus benchmark artifacts: `results/` holds the variant-level and case-level result tables, all gate metrics as JSON, and the 400 Markdown case reports; `data/` holds the byte-locked manifests, case index, and split; `GATES_LOCKED.md`, `AMENDMENT_v1_negative.md`, `PRIOR_ART.md`, `PROGRESS.md`, and `RESUME.md` hold the governance record. Engineering properties that made the evaluation possible: per-variant evidence caches (VEP 27,428; gnomAD 27,428; constraint 2,040 genes; HG002 per-gene slices 2,648) that are resume-safe and never cache a failure as data; threaded API fetchers tuned against Ensembl and gnomAD rate

```
# Case report: PMID_25794864_P1
- Arm: real | Disease: OMIM:117360 | Causal gene: ITPR1
- HPO terms (12): HP:0001251, HP:0001290, HP:0025336, ...
- Panel: 40 genes | Candidates evaluated: 61
| Rank | Gene | Variant | Class | Confidence | Evidence trace |
| 1 | ITPR1 | 3-4645673-C-T | LP | moderate-high | PM2: gnomAD max AF 0.000000 < 5e-5;
PM5: different P/LP missense at same codon (other ClinVar record);
PP3: SIFT=deleterious_low_confidence, PolyPhen=probably_damaging;
PP2: missense in missense-constrained gene ITPR1 (mis_z=7.40>=3.09) |
| 2 | LRRC7 | 1-69883681-T-C | VUS | uncertain | PM2: gnomAD max AF 0.000000 < 5e-5 |
| 6 | GRM1 | 6-146099362-C-T | VUS | uncertain | BS1: gnomAD max AF 0.0250 >= 0.01 |
| 8 | ITPR1 | 3-4820013-C-T | B | very low | BA1: gnomAD max AF 0.5392 >= 0.05 |
Provenance: Ensembl VEP REST, gnomAD v4, ClinVar same-codon (other records only).
```

Figure 8: Excerpt of a real RDCB case report (ranks 3–5, 7, 9–10 elided). The causal variant sits at rank 1 with a four-code evidence trace; the baseline arm ranked it second.

limits; a local GIAB VCF mirror adopted after remote-tabix throttling; and a stage driver that re-derives any artifact from caches in one command. The full pipeline restored from manifest once after a total environment wipe, with every source checksum re-verified (Section 10).

8 Discussion

8.1 Three findings that survived the locked gates

First, the interpretable case report is a benchmarkable artifact, and it can be produced at 100% evidence-trace completeness from public data alone. Prior benchmarks score rankings; RDCB scores the argument. The calibration results say the argument’s confidence language means something in this cohort: stated high and moderate-high confidences were never wrong (78/78 reports), and the uncertain label is right 84.5% of the time — exactly the conservative profile a triage report should have.

Second, phenotype integration is a top-1 intervention, not a top-5 one. The measured tradeoff (+13.4 pp top-1, −1.6 pp top-5 on real cases) is a concrete design result for case-ranking systems. If the product is a single best answer with a trace, integrate phenotype at ranking time; if the product is a shortlist for manual review, rank on variant evidence and show phenotype as context. RDCB’s full arm does the former and its reports show both arms’ ordering.

Third, gene constraint is dangerous as a downgrade criterion for Mendelian LoF variants. The G2 failure mechanism — healthy-carrier tolerance masquerading as mechanism — is quantified here (1,194 of 1,197 downgrades; MCC −0.393). Kept as reported evidence (Amendment 1), constraint is informative without being destructive; every PVS1_Moderate trace in the shipped reports shows the LOEUF datum that would otherwise have silently demoted the call.

8.2 What the negatives cost and buy

Two locked gates failed. The cost is obvious: the headline claim the gates were built to test — phenotype integration improves case-level shortlisting — is not supported. The buy is credibility and information: the top-1 result can be trusted precisely because the headline negative is reported beside it, and the two failures localize *where* integration helps (the top of the list) and where naive constraint use hurts (LoF demotion). A benchmark that only ever passes its gates teaches nothing

about design boundaries.

8.3 Future work

Inheritance-model integration, the component named in the locked gates but not shipped (Section 4.2), is the first item: zygoty-aware ranking against the disease’s inheritance model, implemented behind a new amendment before any new case-level run. Deep-learning comparison arms are a natural fit for this pipeline: AlphaMissense and EVE scores were pre-committed as features-only (never truth) in the scope lock and were not used in this evaluation; adding them as additional ranking features — and as comparison arms against the ACMG arms — would quantify what learned variant effect predictors add to a fully traced report. Extending the background to chrX (when a GRCh38 GIAB chrX benchmark exists), adding SV/CNV scope, and running the benchmark on a prospective undiagnosed cohort with clinician review of the rendered reports would test the artifact where it matters.

9 Limitations

SNV/small-indel scope only; no chrX in simulated backgrounds (GIAB v4.2.1 GRCh38 is autosomal); 123 autosomal panel genes lie outside benchmark-callable regions. Real-arm cases come from published solved case reports, which skew toward exome-amenable disease and clean molecular answers; the simulated arm, while built on a real individual’s background, cannot capture incomplete penetrance, mosaicism, or regulatory variation. The pipeline’s VUS-heavy causal class distribution reflects strict evidence re-derivation without clinical correlate data (segregation, functional assays, de novo status) — the reports are as strong as public evidence allows, no stronger. The inheritance/zygoty-fit component named in the locked gates is not implemented in the shipped arm (Section 4.2). Ensembl and gnomAD API drift means exact numbers are tied to the byte-locked manifests and cached evidence, all of which ship with the benchmark; gate definitions, not specific API responses, are the durable artifact.

10 Reproducibility

Repository layout: `code/`, `data/` (manifests + SHA-256 checksums), `results/`, `paper/`. `GATES_LOCKED.md` (commit 134a6ae) predates every results commit; `AMENDMENT_v1_negative.md` predates every case-level result; the full history is public. `RESUME.md` documents a complete environment restore: one full restore was performed after a sandbox wipe and every source checksum re-verified against the byte-locked manifest. `MANIFEST.sha256` covers the shipped payload (438 files) with all entries verified. All external data are public; no wet-lab work; no proprietary features; no result was computed from bytes that failed verification.

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